

The apparel retailer sells clothing, jewelry, and footwear items directly to customers through its web-based portal. The products supplied by its multiple vendors are first packed in cases. Multiple cases are then aggregated and assembled into pallets for delivery to the retailer's main warehouse. After pallets arrive at the warehouse, they are verified against a purchase order and inspected. Once the verification and inspection processes are completed, warehouse personnel at the receiving department break down the pallets, put away the items into their corresponding storage locations, and update their inventory records. The retailer's customers place orders online and items are picked by warehouse personnel, packed, and shipped to the customers.

The scope of the WMS is limited to operations happening with the warehouse. It starts with shipments received from vendors based on purchase orders submitted by the retailer. The database captures the details of shipments which are a realization of purchase orders sent to the vendors. The preparation and submission of the actual purchase orders based on the stock on hand and inventory policies is handled by another subsystem and is outside the scope of the WMS. The WMS database stores data that is needed to support the following warehouse operations - receiving items from vendors to replenish on-hand inventory, the stocking of items, and the picking of items to fulfil customer orders. The database stores vast amounts of information related to customers, vendors, customer orders, shipments, pallets, cases, items, etc.

The following descriptions should give you an idea of the type of information that is stored.

Vendor: Several vendors supply the items that apparel retailer sells through its web-based portal. A vendor may have zero to many active purchase orders, but a purchase order is always associated with a single vendor. A vendor has an identifier (vendor number), name, address, phone number, and email address.

Purchase Order and Shipment: The retailer orders items from its vendors via purchase orders. A vendor shipment may be composed of many pallets, and a single pallet may contain many cases. The warehouse accepts mixed pallets from its vendors (i.e., a pallet may contain cases of different types of items), but a case must contain a single type of item. A vendor shipment has an identifier, receipt date, number of pallets, and item information (e.g., the identifiers of the item types included in the pallets).

Pallet: Pallets are used by vendors to ship cases of items to the retailer's warehouse. These pallets, which are owned by apparel retailer, are made of a durable composite material, and can be individually tracked by means of a radio frequency identification (RFID) tag. The retailer owns several thousand pallets, which it distributes among its vendors so that they can fulfill orders. A pallet has an identifier (pallet ID), weight, volume, and case quantity (when loaded).

Case: A case is used to ship a single type of item (e.g., shirts or pants). Cases are also owned by the retailer and can be individually tracked by means of an RFID tag. The retailer owns several thousand cases, which it distributes among their vendors so that they can fulfill orders. A case has an identifier (case ID), item number, item quantity, volume, and weight.

Inventory Location: Pallets are delivered by vendors to the retailer's main warehouse. Once a shipment arrives at the loading dock, the truck driver presents shipping papers (e.g., a Bill of Lading and a packing list) to the warehouse personnel in the receiving area. The warehouse personnel verify the shipment contents against a purchase order to ensure that the right items and the right quantities are being delivered. If everything is in order, warehouse personnel will break down the pallets and store the items contained in the cases in one or many of the thousands of inventory locations within the warehouse. An inventory location has an identifier (location number).

Customer Order: Customers order items from the apparel retailer via a customer order. Fulfilling customer orders depletes the retailer's inventory. A customer may order many different types of items via a customer order. In addition, a customer may have zero to many orders in process at any time. A customer order has an identifier (order number), order date, requested ship date, actual ship date, shipping carrier, number of items, item numbers, and item quantities.

Item: The apparel retailer sells many different types of items to customers including belts, footwear, jewelry, neckwear, outerwear, shirts, and slacks. An item has an identifier (item number), description, price, weight, annual sales, and stock location.

Customer: Customers order items from the apparel retailer through their web-based sales portal. A customer may have several active orders, but there may also be customers with no active orders. A customer has an identifier (customer number), name, address, phone number, and email address.

Additional notes

- Items included on a pallet could sometimes represent a consolidation of items ordered on multiple purchase orders sent to a vendor.
- Each item has a designated storage location, and a location is typically shared by several items to improve space utilization.
- The weights and volumes of pallets and cases are for empty pallets and cases and are stored in the database to support a future need.
- An item when it is received is in a case on a particular pallet and is related to a purchase order. When it is put away, it is an item in storage in a particular stock location, and when it is picked it is an item related to a particular customer order.